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UNIT III | Chapter 8

Excretion

Chapter Outline

8.1 Modes of Excretion
8.2 Human excretory system
8.3 Mechanism of urine formation in human
8.4 Regulation of kidney function
8.5 Micturition
8.6 Role of other organs in excretion
8.7 Disorders related to the excretory system
8.8 Haemodialysis

Seabirds have no problem in drinking sea water

Learning Objectives:

- *Understands different modes of excretion in animals.*
- *Learns the structure of the human excretory system.*
- *Understands the structure of a nephron, mechanism of urine formation - glomerular filtration reabsorption and secretion from the renal tubules.*
- *Visualizes the blood supply to the kidney including the nephrons*
- *Learns about the possible kidney related diseases.*

6MERK5

Earliest animal life forms arose around 700 million years ago. They were marine organisms like the modern sponges. Each cell of a modern sponge is surrounded by sea water, but it maintains an intracellular ionic composition different from that of the sea water. Evolution led to changes in the organisation of the tissue layers followed by formation of specialized external tissue layers. This provided a barrier between the external environment and internal fluid resulting in the formation of extracellular fluid. Major changes in osmoregulation and ionic regulation occurred during the evolution of chordates. The ability to control extracellular fluid composition was essential for the diversification of animals to inhabit brackish water, fresh water and land. Animals that invaded land had the risk of desiccation and were unable to excrete metabolic waste directly into the water; hence there was a need for an alternate pathway to dispose the nitrogenous wastes.

Most animals rely on kidneys to control ionic and water balance. Some